7CS039 Lab 5 Applications I

1 Objectives

By the end of this session we will be able to

- Conduct Chi Squared Test of Independence using `R`.
- Interpret the results.

2 Chi Squared Testing for the Independence of Two Variables

In this weeks lecture and tutorial we discussed Chi Squared Testing for the Independence of Two Variables. We had the following example.

```
> DrivingTest<-matrix(c(24,56,33,11,40,24),ncol=3,byrow=TRUE)
> colnames(DrivingTest) <- c("A","B","C")
> rownames(DrivingTest) <- c("pass", "fail")
> DrivingTest<-as.table(DrivingTest)
> DrivingTest
```

```
> DrivingTest<-matrix(c(24,56,33,11,40,24),ncol=3,byrow=TRUE)
> colnames(DrivingTest) <- c("A","B","C")
> rownames(DrivingTest) <- c("pass", "fail")
> DrivingTest<-as.table(DrivingTest)
> DrivingTest
      A  B  C
pass 24 56 33
fail 11 40 24
>
```

The table illustrates the numbers of people that passed or failed driving tests administered by three different examiners A,B , and C. The obvious question to ask is: "Is the outcome of the test dependent (or contingent) on the examiner giving the test" Thus we have the null and alternative hypothesis below

- H_0 the outcome is independent of the tester.

- H_1 the outcome depends on the tester.

Remember, that the key point here is that we are interested in the difference between what we have observed (i.e. the numbers in the table) and what we would expect to observe if the null hypothesis is true. To conduct a chi squared test for independence we do

```
> summary(DrivingTest)
```

```
> summary(DrivingTest)
Number of cases in table: 188
Number of factors: 2
Test for independence of all factors:
        Chisq = 1.2881, df = 2, p-value = 0.5252
>
```

The observed value of our test statistic is

$$\chi^2_{obs} = 1.2881$$

By default we are testing at the $\alpha = 0.05$ level of significance. Our p value is $p = 0.5252$ which is definitely not less that $\alpha = 0.05$ so we do not reject H_0. We cannot claim that the choice of tester has any bearing on the result of a persons driving test.

3 Anaphylaxis After Receipt of the First Dose of Pfizer-BioNTech COVID-19 Vaccine

Anaphylaxis is a severe allergic reaction to venom, food, or medication. Most cases are caused by a bee sting or eating foods that are known to cause allergies, such as peanuts or tree nuts. Anaphylaxis causes a series of symptoms, including a rash, low pulse, and shock, which is known as anaphylactic shock. On January 21st 2021 a study was published by the CDC into incidences of Anaphylaxis amongst recipients of the first does of the Pfizer-BioNTech covid-19 vaccine. The following data is taken directly from the paper below

Shimabukuro T, Nair N. Allergic Reactions Including Anaphylaxis After Receipt of the First Dose of Pfizer-BioNTech COVID-19 Vaccine. JAMA. Published online January 21, 2021. doi:10.1001/jama.2021.0600

	Male	Female	Total
(Anaphylaxis) Yes	2	19	21
(Anaphylaxis) No	648325	1177508	1825833
Total	648327	1177527	1825854

From the table there seems to be a roughly $1/3$ to $2/3$ split the numbers of men and women given the vaccine respectively but that split is not reflected in the numbers of patients experiencing anaphylaxis. Is there a link between anaphylaxis and gender? Are females more likely to experience anaphylaxis than males or is the experiencing of anaphylaxis independent of gender? We can carry out a test of independence to see. First we construct the contingency table.

```
> Allergy<-matrix(c(2,19,648325,1177508),ncol=2,byrow=TRUE)
> colnames(Allergy) <- c("Male","Female")
> rownames(Allergy) <- c("(Anaphylaxis) Yes", "(Anaphylaxis) No")
> Allergy<-as.table(Allergy)
> Allergy
```

```
> Allergy<-matrix(c(2,19,648325,1177508),ncol=2,byrow=TRUE)
> colnames(Allergy) <- c("Male","Female")
> rownames(Allergy) <- c("(Anaphylaxis) Yes", "(Anaphylaxis) No")
> Allergy<-as.table(Allergy)
> Allergy
                     Male  Female
(Anaphylaxis) Yes       2      19
(Anaphylaxis) No   648325 1177508
>
```

Now we state our null and alternative hypothesis.

- H_0 the outcome (anaphylaxis) is independent of gender.
- H_1 the outcome (anaphylaxis) depends on gender.

```
> summary(Allergy)
```

```
> summary(Allergy)
Number of cases in table: 1825854
Number of factors: 2
Test for independence of all factors:
        Chisq = 6.192, df = 1, p-value = 0.01283
>
```

The p-value is less than $\alpha = 0.05$ therefore we reject the null hypothesis and conclude, that based on this data, there is evidence that experiencing anaphylaxis (after taking the first does of the Pfizer-BioNTech vaccine) is dependent on gender.

It is worth noting that the chances of experiencing anaphylaxis are extremely small (approximately 21 in 1.9 million) but the difference between the number of cases in men and women is statistically significant.

Activity Time

In December 2020 the below article was published by the National library of Medicine.

Spassiani I, Gubian L, Palu G, Sebastiani G. Vaccination Criteria Based on Factors Influencing COVID-19 Diffusion and Mortality. Vaccines (Basel). 2020 Dec 15;8(4):766. doi: 10.3390/vaccines8040766. PMID: 33334007; PMCID: PMC7765372.

This paper analyses public data from 19000 patients from the period February to May 2020 and looks for relationships between various comorbidities in an effort to prioritize vaccination. The paper is available here
https://www.mdpi.com/2076-393X/8/4/766
Contingency tables based on their data are provided in an accompanying document here
http://www.mdpi.com/2076-393X/8/4/766/s1
Manually recreate these contingency tables in `R` and test for independence between the relevant variables.